

Workshop on Advanced Sampling Methodologies

Introduction to R Programming

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Workshop's agenda

- **Day 1:** Introduction to QGIS
- **Day 2:** Automatic update of census EA and national sampling frame
- **Day 3:** Introduction to R programming
- **Day 4:** Introduction to Sampling Design
- **Day 5:** Advanced R programming
- **Day 6:** Earth observation (EO) for sampling design
- **Day 7:** AI and machine learning for sample selection
- **Day 8:** Simulation and optimization of sampling strategies
- **Day 9:** R Shiny for sampling design evaluation
- **Day 10:** Wrap-up and workshop evaluation

Today's agenda

- Introduction to R and RStudio
- Data types and data structures
- Working with scripts and the console
- Importing data
- Saving and closing sessions
- Practical exercises

What is R?

Why R?

- Powerful tool for **data analysis and visualization**
- Encourages **reproducibility** and **sharing**
- Thousands of user-created **packages** extend its functionality
- Large and active **community** with extensive support

Iris k-means clustering

X Variable

Sepal.Length

Y Variable

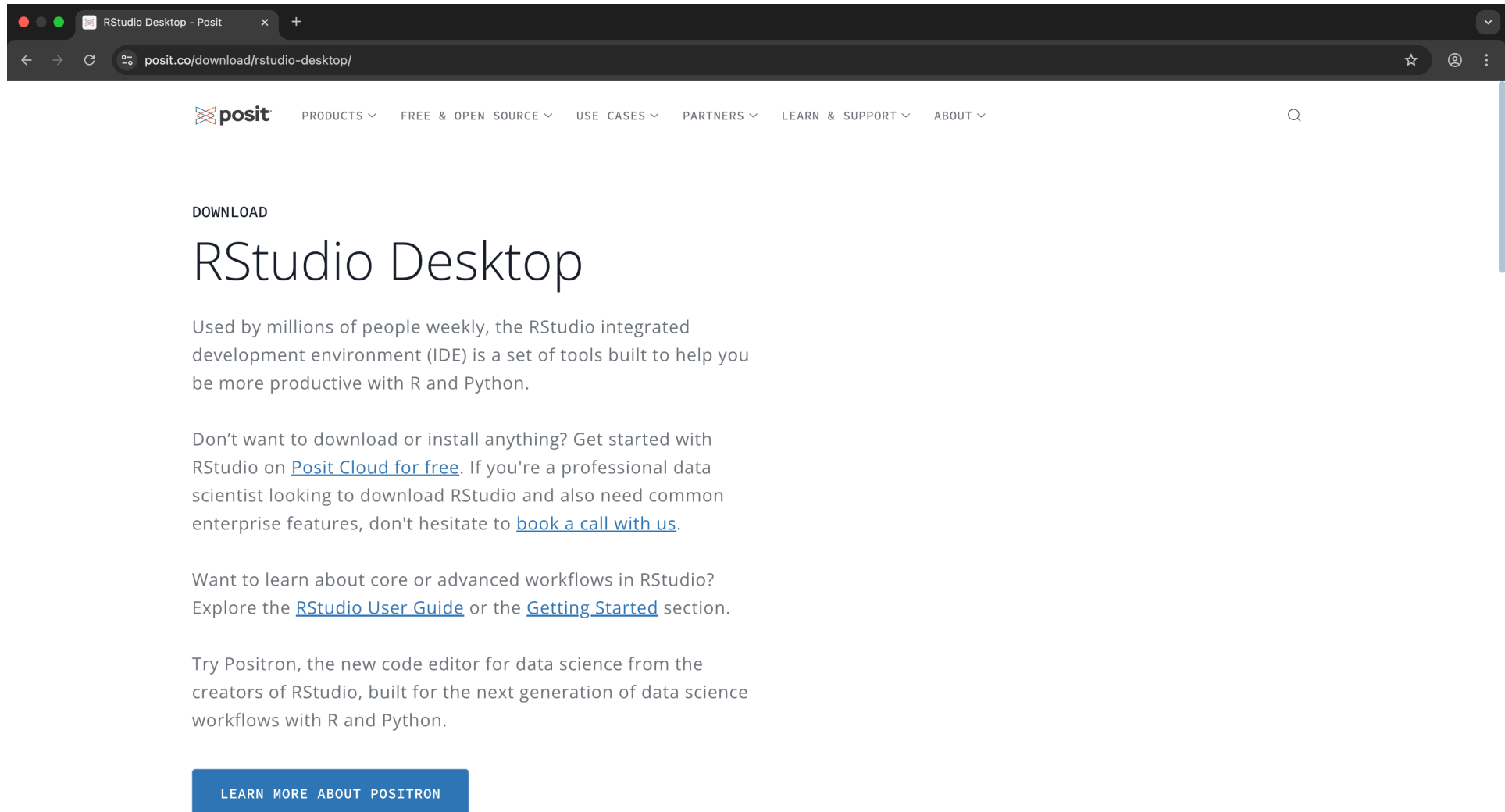
Sepal.Width

Cluster count

3

Installing R and RStudio

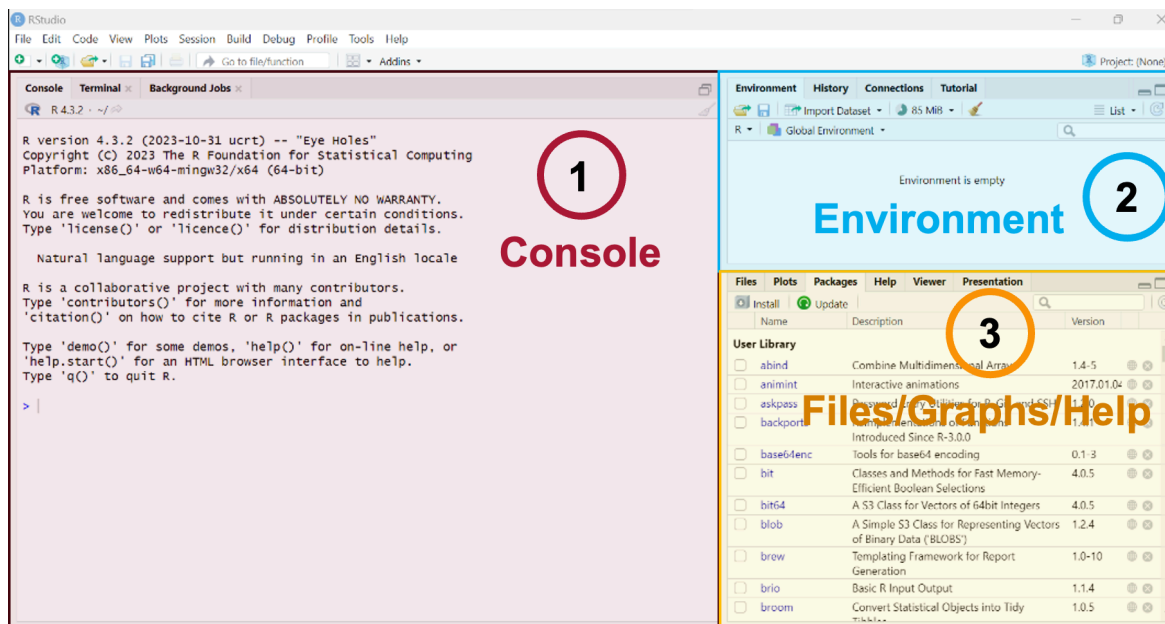
- Download and install **R and RStudio** from posit.co



RStudio interface

RStudio has four main panes:

- **Source Editor** – write and save code
- **Console** – run commands interactively
- **Environment/History** – see and manage objects
- **Files/Plots/Packages/Help/Viewer** – tools and outputs



 Try to navigate through the different panes.

Script Editor vs Console

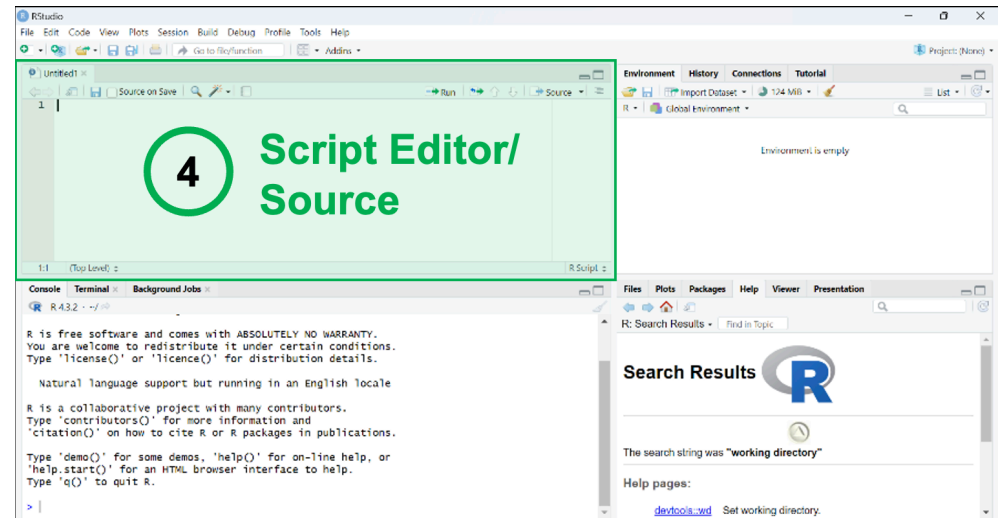
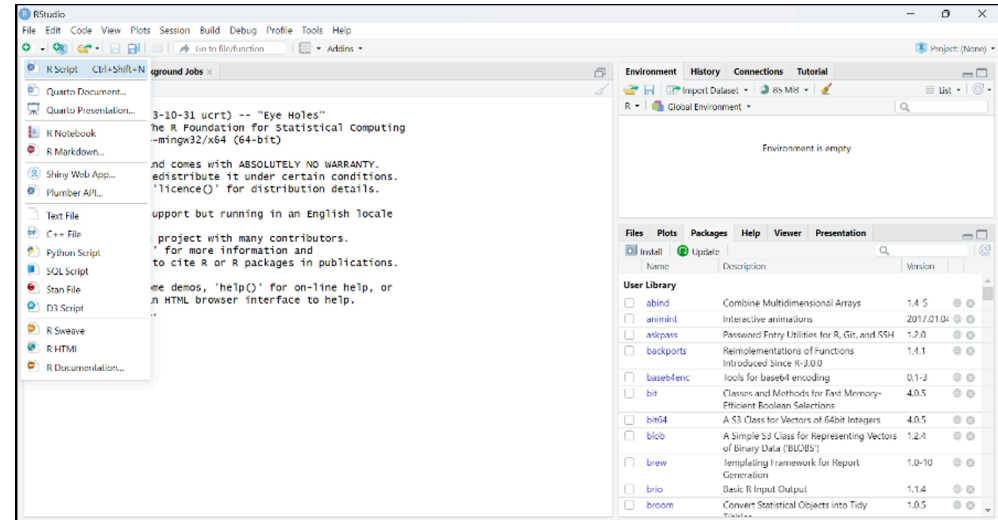
Script Editor

- Write, edit, and save code
- Organize analysis in reproducible scripts
- Prevent accidental execution before editing

Console

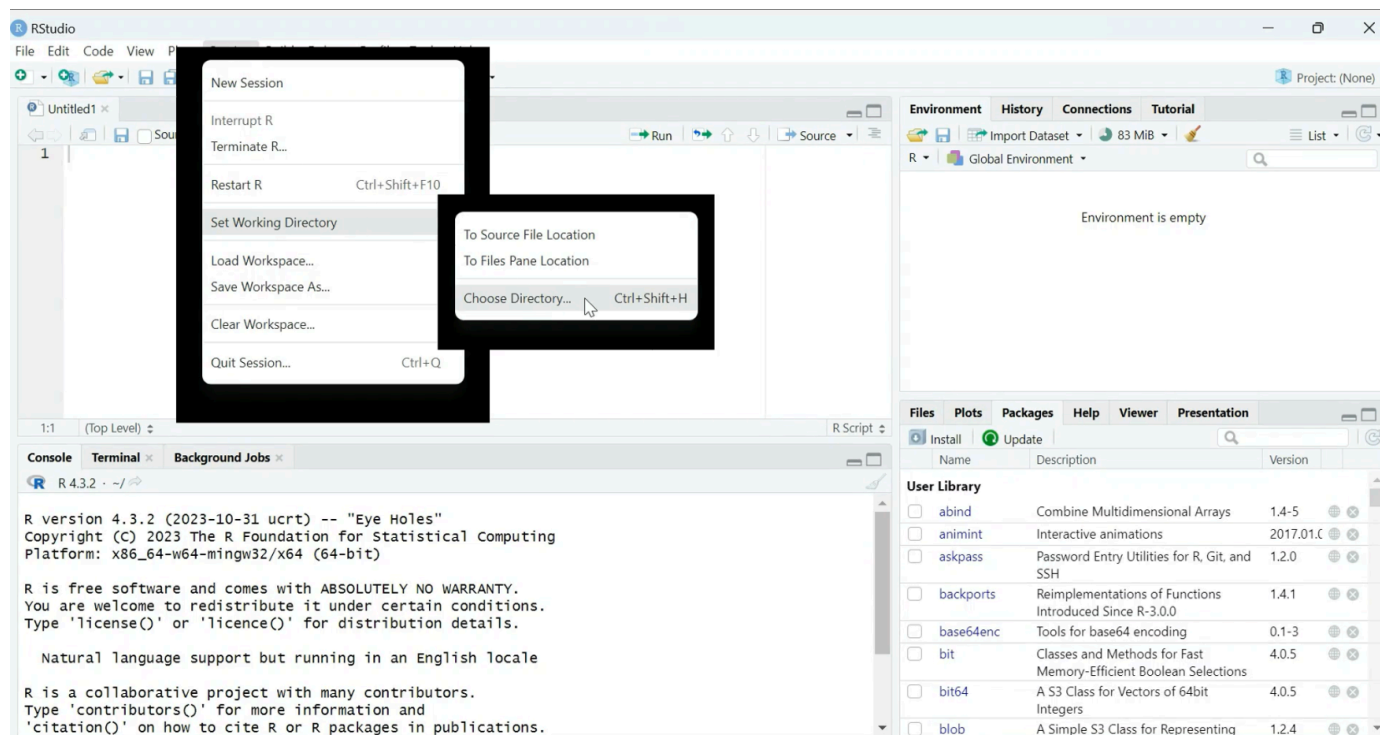
- Executes code immediately
- Use for quick checks and interactive testing
- Output and errors appear here

 Best practice: write in the Script Editor, run via Console.



Working Directory

- Your working directory is the folder where R looks for and saves files.
- Set with: `setwd("path")`
- Check with: `getwd()`
- Good practice: create a dedicated folder for each project.



 Try to run the code in your R Console and in the Script Editor.

R as a calculator

- R can perform arithmetic directly

```
1 1 + 2 * 8
2 a <- log(10)
3 b <- 5
4 b + b
```

- Results in Console are prefixed with `[1]` (index of first element).
- `[2]`, `[3]`, `...` appear if output spans multiple lines.

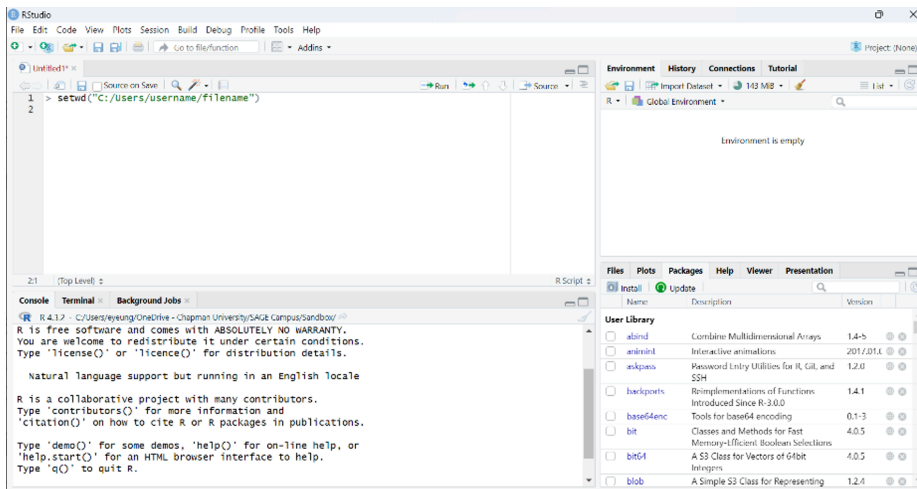
 Try to run the code in your R Console and in the Script Editor.

R packages

- Collections of functions, datasets, and documentation.
- Expand R beyond base functions.
- Install from CRAN, GitHub, or other repositories.

In RStudio

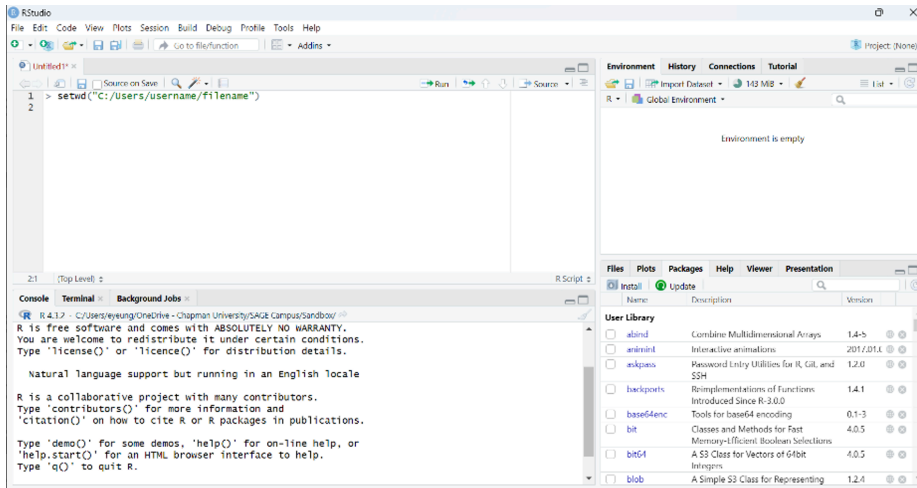
- Use the *Packages* tab to install, update, or remove.
- Or install in the Script Editor



1 `install.packages("dplyr")`

Loading packages

- Load packages into a session via checkbox in the *Packages* pane.



- Or load packages into a session with:

```
1 library(dplyr)
```

 Best practice: only load what you need.

R objects

- Everything in R is an **object** (vectors, lists, functions, data frames, etc.).
- Assign values using `<-` (**Alt** + `-` (minus sign) on Windows and Linux, and **Option** + `-` on a Mac).

```
1 x <- 5
2 x + x    # returns 10
```

Naming tips

-  Descriptive and concise
-  Use underscores `_` or periods `.`
-  Avoid very long names
-  Avoid existing function names
-  No blank spaces

 Try to run the code in your Script Editor.

Data types

- **Numeric:** numbers, may include decimals.

```
1 w <- c(1.7, 3)
2 x <- c(1, 2, 3) # integers are numeric without decimals
```

- **Character:** text strings.

```
1 y <- c("male", "female")
```

- **Logical:** TRUE/FALSE values.

```
1 y <- c(TRUE, FALSE)
```

- **Factor:** categorical variables.

```
1 gender <- factor(c("boy", "girl", "girl", "boy"))
```

Check type:

```
1 class(object)
2 typeof(object)
```

Data structures

- **Vectors:** one-dimensional, homogeneous elements
- **Matrices:** two-dimensional, homogeneous elements
- **Lists:** flexible, heterogeneous elements
- **Arrays:** multi-dimensional, homogeneous elements
- **Data frames:** tabular, columns may differ in type

Vectors

- The most basic structure in R
- One-dimensional, elements must be the **same type**

```
1 numbers <- c(1, 2, 3, 4, 5)
2 colors <- c("red", "green", "blue")
3 logical_values <- c(TRUE, FALSE, TRUE)
```

Indexing with brackets:

```
1 numbers[1]      # first element
2 colors[2:3]     # second and third elements
```

Vectorized operations apply to all elements:

```
1 numbers * 2     # multiplies each element by 2
```

 Try to run the code in your Script Editor.

Matrices

- Two-dimensional, **homogeneous elements**
- Create with `matrix()`

```
1 rownames <- c("math", "science", "history", "English", "law")
2 colnames <- c("male", "female")
3 subject_matrix <- matrix(
4   c(10,9,11,13,12,18,17,13,6,5),
5   nrow = 5, ncol = 2, byrow = TRUE,
6   dimnames = list(rownames, colnames)
7 )
```

- Add row: `rbind()`
- Add column: `cbind()`

 Try to add a row to `subject_matrix` for `geography <- c(4,2)` using `rbind` in your Script Editor.

Matrices to Data Frames

- Convert a matrix into a data frame to **mix numeric and non-numeric types**.

```
1 matrix_stats <- matrix(  
2   c(4015, 3980, 3756, 3333, 4000, 3000, 2000, 1000),  
3   nrow = 4, ncol = 2, byrow = TRUE,  
4   dimnames = list(  
5     c("freshmen", "sophomores", "juniors", "seniors"),  
6     c("enrollment", "living on campus")  
7   )  
8 )  
9  
10 df_stats <- as.data.frame(matrix_stats)  
11 avg_grades <- c("A-", "B+", "B-", "A+")  
12 school_stats <- cbind(df_stats, avg_grades)
```

 Try to run the code in your Script Editor.

Data Frames

- Tabular structure with columns of different types
- Similar to spreadsheets

```
1 cats <- data.frame(  
2   coat = c("Persian", "black", "tabby"),  
3   weight = c(2.1, 5.0, 3.2),  
4   likes_string = c(TRUE, FALSE, TRUE)  
5 )  
6  
7 animal_sleep <- data.frame(  
8   ranking = c(4, 1, 2, 3),  
9   animal = c("koala", "capybara", "camel", "panda"),  
10  country = c("Australia", "Brazil", "Egypt", "China"),  
11  avg_sleep_hr = c(20, 3, 6, 12)  
12 )
```

 Try to run the code in your Script Editor.

Lists

- Flexible data structure with elements of **different types**
- Can include vectors, data frames, matrices, or other lists

```
1 my_list <- list(  
2   name = "Alice",  
3   age = 25,  
4   scores = c(90, 85, 92),  
5   passed = TRUE  
6 )  
7  
8 my_list$name  
9 my_list[[2]]
```

 Try to run the code in your Script Editor.

Arrays

- Extend vectors to two or more dimensions
- All elements must be the same type

```
1 arr <- array(1:12, dim = c(2, 3, 2))  
2 arr  
3 arr[1, 2, 1] # element at row 1, col 2, slice 1
```

Useful for structured multi-dimensional data. However, this is hardly used in most cases.

 Try to run the code in your Script Editor.

Importing data

- Create data manually, but usually import from files

Common formats

- **Text:** `.txt` (`readLines()` function)
- **Tabular data:** `.csv`, `.tsv` (`read.table()` function or `read_csv()` function from the `readr` package)
- **Excel:** `.xlsx` (`xlsx` package)
- **Google sheets:** (`googlesheets` package)
- **Statistics program:** SPSS, SAS (`haven` package)
- **Databases:** MySQL (`RMySQL` package)

Example:

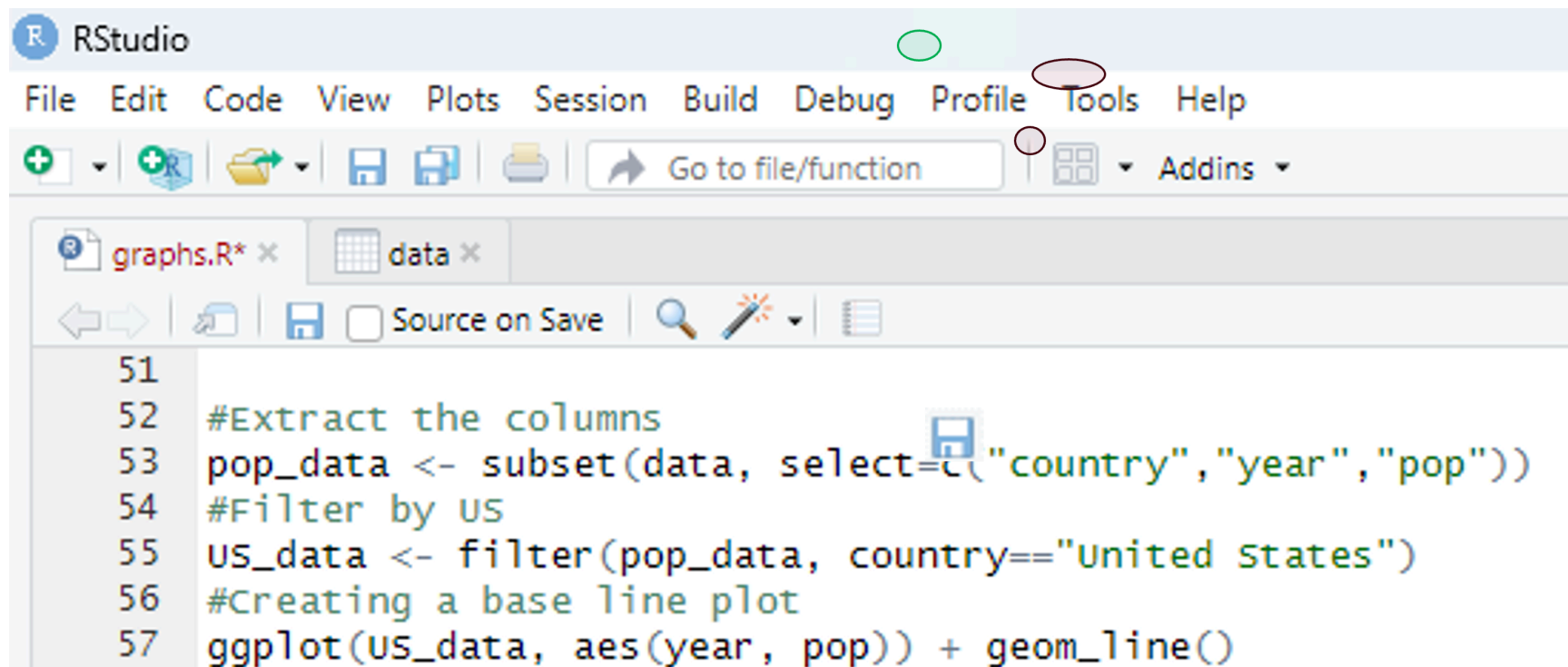
```
1 csv_data <- read_csv("data/world_data.csv")
```

➡ Always use forward slashes `/` in file paths.

i Try to run the code in your Script Editor.

Saving your script

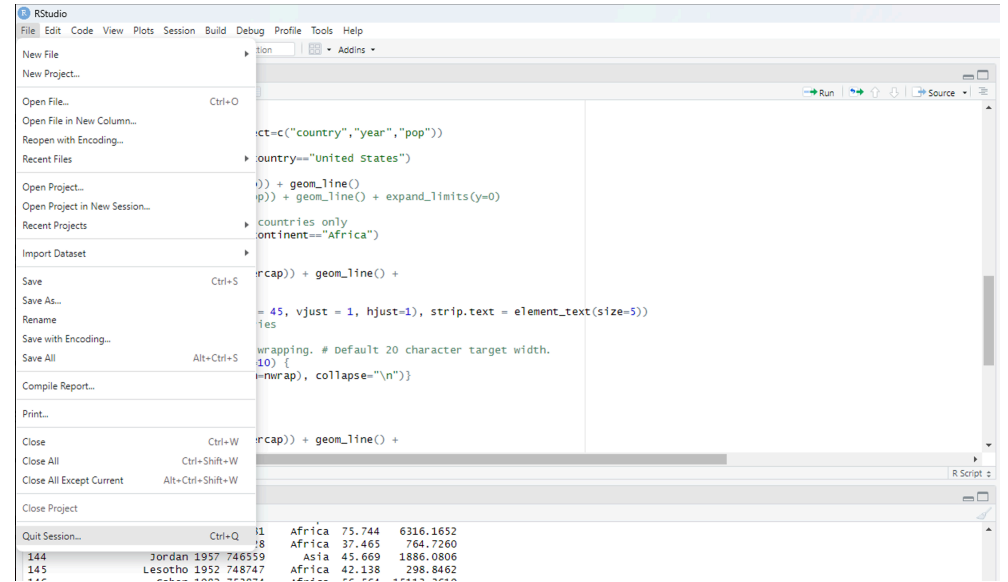
- Save all steps of analysis in `.R` scripts
- Add comments with `#`
- Save with **File** → **Save** or `Ctrl+S` / `Cmd+S`
- Unsaved scripts show in red with an asterisk `*`



Closing an R Session

Options:

- **File** → **Quit Session**
- `q()` in Console



- ➡ Do **not** save the workspace image. Always start fresh for reproducibility.
- i Try to close and reopen your R Session.

R cheatsheets

- R and RStudio are relatively difficult to learn at the beginning
- Once you master the basis it is much easier to move forward
- R cheatsheets are an excellent resource for beginners

Base R Cheat Sheet

Getting Help

Accessing the help files

?mean
Get help of a particular function.
help.search('weighted mean')
Search the help files for a word or phrase.
help(package = 'dplyr')
Find help for a package.

More about an object

str(iris)
Get a summary of an object's structure.
class(iris)
Find the class an object belongs to.

Using Packages

install.packages('dplyr')
Download and install a package from CRAN.

library(dplyr)
Load the package into the session, making all its functions available to use.

dplyr::select
Use a particular function from a package.

data(iris)
Load a built-in dataset into the environment.

Working Directory

getwd()
Find the current working directory (where inputs are found and outputs are sent).

setwd('C://file/path')
Change the current working directory.

Use projects in RStudio to set the working directory to the folder you are working in.

Vectors

Creating Vectors

<code>c(2, 4, 6)</code>	<code>2 4 6</code>	Join elements into a vector
<code>2:6</code>	<code>2 3 4 5 6</code>	An integer sequence
<code>seq(2, 3, by=0.5)</code>	<code>2.0 2.5 3.0</code>	A complex sequence
<code>rep(1:2, times=3)</code>	<code>1 2 1 2 1 2</code>	Repeat a vector
<code>rep(1:2, each=3)</code>	<code>1 1 1 2 2 2</code>	Repeat elements of a vector

Vector Functions

sort(x) Return x sorted.	rev(x) Return x reversed.
table(x) See counts of values.	unique(x) See unique values.

Selecting Vector Elements

By Position	
<code>x[4]</code>	The fourth element.
<code>x[-4]</code>	All but the fourth.
<code>x[2:4]</code>	Elements two to four.
<code>x[-(2:4)]</code>	All elements except two to four.
<code>x[c(1, 5)]</code>	Elements one and five.
By Value	
<code>x[x == 10]</code>	Elements which are equal to 10.
<code>x[x < 0]</code>	All elements less than zero.
<code>x[x %in% c(1, 2, 5)]</code>	Elements in the set 1, 2, 5.
Named Vectors	
<code>x['apple']</code>	Element with name 'apple'.

Programming

For Loop

```
for (variable in sequence){  
  Do something  
}
```

Example

```
for (i in 1:4){  
  j <- i + 10  
  print(j)  
}
```

While Loop

```
while (condition){  
  Do something  
}
```

Example

```
while (i < 5){  
  print(i)  
  i <- i + 1  
}
```

If Statements

```
if (condition){  
  Do something  
} else {  
  Do something different  
}
```

Example

```
if (i > 3){  
  print('Yes')  
} else {  
  print('No')  
}
```

Functions

```
function_name <- function(var){  
  Do something  
  return(new_variable)  
}
```

Example

```
square <- function(x){  
  squared <- x*x  
  return(squared)  
}
```

Reading and Writing Data

Also see the [readr](#) package.

Input	Output	Description
<code>df <- read.table('file.txt')</code>	<code>write.table(df, 'file.txt')</code>	Read and write a delimited text file.
<code>df <- read.csv('file.csv')</code>	<code>write.csv(df, 'file.csv')</code>	Read and write a comma separated value file. This is a special case of read table/ write table.
<code>load('file.Rdata')</code>	<code>save(df, file = 'file.Rdata')</code>	Read and write an R data file, a file type special for R.

Conditions	a == b	Are equal	a > b	Greater than	a >= b	Greater than or equal to	is.na(a)	Is missing
	a != b	Not equal	a < b	Less than	a <= b	Less than or equal to	is.null(a)	Is null

Take home

- R is a **versatile tool for data analysis and visualization** – free, open-source, and supported by a vast community with thousands of packages.
- RStudio **enhances productivity** – organize, edit, and execute code efficiently using the Script Editor, Console, and built-in panes.
- Understand **R objects, data types, and structures** – Vectors, matrices, lists, data frames, and arrays form the building blocks of R programming.
- **Reproducibility** matters – save scripts, comment your code, and avoid saving workspace images to ensure analyses can be repeated.
- Packages **expand functionality** – install, load, and use packages to extend R's capabilities for importing, manipulating, and visualizing data.

Resources

R Core Team. (2025). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org>

The official reference for R itself, including installation, base functions, and documentation.

RStudio Team. (2025). RStudio: Integrated Development Environment for R. <https://posit.co>

Wickham, H., & Grolemund, G. (2017). R for Data Science: Import, Tidy, Transform, Visualize, and Model Data. O'Reilly Media. <https://r4ds.had.co.nz>

Venables, W. N., Smith, D. M., & the R Core Team. (2023). An Introduction to R. CRAN. <https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf>

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning with Applications in R (2nd edition). Springer.

Exercise

Please download the R script with exercises from **GitHub** and try to complete it.